

The diagnostic pathway and prognosis in bulbar-onset amyotrophic lateral sclerosis

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ABSTRACT

Background: Despite the inevitability of disease progression in amyotrophic lateral sclerosis, there is a high degree of prognostic heterogeneity in all subtypes. Some bulbar-onset (BO) patients may develop rapid anarthria yet remain ambulant for a prolonged period, whereas others progress rapidly, with early generalisation of motor weakness to the limbs and respiratory muscles. Diagnostic delay is a common occurrence in ALS, and many BO patients report having attended other specialist clinics prior to diagnosis. **Methods:** A retrospective descriptive study of BO ALS patients seen in a tertiary clinic over a six year period. **Results:** Forty-nine BO ALS patients were studied. Median survival from symptom onset was 27 months (range 6–84). 63% of subjects were female and the mean age at symptom onset was 68 years. Half had been referred to another speciality prior to diagnosis, either otolaryngology or stroke clinics, but this did not influence diagnostic latency or survival. Emotionality was reported in 45% of patients. Neurophysiological assessment was performed in 80%, brain imaging recorded in 69%, and antibody testing for myasthenia gravis in 22%. The median time to symptomatic progression beyond the bulbar region was approximately 1 year, with equal proportions progressing to the upper or lower limbs. The median interval from onset to anarthria was 18 months, and to loss of ambulation 22 months. There was a close correlation between the two ($r^2 = 0.6$) and median survival from loss of ambulation was only 3 months. Gastrostomy was carried out in 78% of patients with a median time of 13 months from symptom onset, and 3 months from diagnosis. Median survival from gastrostomy was 10 months. **Conclusions:** Survival in bulbar-onset ALS is highly variable. Half of the patients were referred to an inappropriate clinic prior to diagnosis. The time interval to the development of anarthria predicted the timing of subsequent loss of ambulation accurately from which survival may then be only a few months.

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1. Introduction

Understanding the clinical heterogeneity inherent in amyotrophic lateral sclerosis (ALS) remains an important challenge that might have some bearing on the disappointing results from clinical therapeutic trials for this degenerative disorder of cerebral and spinal motor neurons. Despite a median survival of 2–3 years from symptom onset in most clinic-based series, there is a skewed distribution with a significant number surviving beyond 10 years, a small minority of which also have bulbar-onset (BO) of their symptoms [1]. BO patients constitute up to 25% of most ALS clinic populations, and generally this phenotype is associated with more rapid progression [2]. It has been observed that older female patients are over-represented in BO patients [3–5], which links to the independent observation that the female:male ratio in ALS increases with each decade [5–7].

The insidious onset of symptoms in ALS, the lack of a diagnostic test, erroneous referral to other specialists [8], and even unnecessary surgery

in some cases [9], are all factors involved in the consistent delay from symptom onset to formal diagnosis in ALS (frequently beyond 12 months [2]). This interval might represent a missed opportunity for a more effective therapeutic intervention. True ‘mimics’ of ALS are rare, possibly even more so for BO patients, who are observed to reach ALS clinics sooner after symptom onset than limb-onset patients [10], though this may simply reflect more rapid progression of symptoms. We have informally observed that BO patients are frequently referred to otolaryngology and stroke clinics prior to diagnosis in our tertiary centre, sometimes undergoing unnecessary investigations as a result.

Finally, we also recognise a group of patients, often female, who despite the relatively rapid development of anarthria, remain ambulant for months thereafter, and may derive proportionately a greater benefit from interventions such as enteral feeding.

In this retrospective analysis we sought to characterise the diagnostic pathway and aspects of clinical heterogeneity affecting survival in bulbar-onset ALS patients.

2. Methods

Patients with ALS seen at a tertiary referral centre since August 2003 were prospectively recorded in a database as part of routine clinical care. Since 2009 attendees have provided written informed

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consent for the use of anonymised clinical data for research and publication, and approval for the retrospective use of deceased patients (or those lost to follow up) was granted through application to the Ireland Health and Social Care Research Ethics Committee 2 (09/NIR02/35).

The diagnosis of ALS was made by consultant neurologists with extensive experience (KT, MT), after exclusion of other conditions, and according to revised El Escorial criteria [11].

The database included clinical data recorded prospectively at regular three-monthly clinic attendances, and was used in conjunction with a thorough retrospective analysis of the clinical notes and correspondence. We considered several potential prognostic factors that could be established with confidence from the clinical records: gender, age at symptom onset, diagnostic latency, prior specialist referral, progression of symptoms beyond the bulbar territory, the development of anarthria (loss of all useful speech), loss of ambulation, emotionality, neurological investigations prior to diagnosis, gastrostomy insertion, and any use of riluzole.

Progression of symptoms beyond the bulbar territory was defined by either patient-reported limb weakness or objective weakness or wasting of limbs noted on examination. Diagnostic latency was defined as the time from symptom onset to diagnosis at our tertiary centre.

Survival was calculated as the time from symptom onset to death or census on 1 March 2009. Survival status was ascertained through GP contact for those lost to follow up.

Independence of categorical variables was tested using the Chi square test, with ANOVA testing between categorical and continuous variables, and linear regression with ANOVA for comparison between continuous variables. Median survival analysis was performed using univariate Kaplan–Meier (K–M) log-rank, with continuous variables categorised above and below the mean. Calculations were performed using SPSS (version 14, SPSS Inc., Chicago, Illinois).

Direct comparison with other ALS phenotypes was not a primary objective of this study due to incomplete data. However, we made use of a previously complete dataset from a published study of lower limb-onset ALS patients derived from our clinic [12], using it to

Table 1

Characteristics and survival of bulbar-onset patients (all in months unless stated).

Category	Variable	Bulbar-onset	Lower limb-onset [12]
Survival	Median (95% CI)	27 (20–34)	41 (33–49)
	Mean (95% CI)	33 (27–39)	73 (53–92)
	Range	6–84	5–285
Demographics	Mean age of onset in years (SD, range)	68 (11, 43–90)	60 (11, 33–84)
	Gender ratio F:M	1.7:1	1:1
Diagnostic pathway	Mean diagnostic latency (median, SD, range)	10 (9, 6, 2–34)	22 (15, 23, 2–216)
	Diagnostic latency versus survival	$r^2 = 0.144$ ($p = 0.008$)	
	K–M median	33 versus 23 ($p = 0.012$)	
	Referral to other clinic	49%	
		Otolaryngology 54%	
		Stroke 42%	
Progression	Source of referral to tertiary clinic	Neurologist 74%	
		GPs 14%	
		Stroke physicians 10%	
	Beyond bulbar region at census	59%	
		Upper limb 50%	
		Lower limb 50%	
	Median interval onset to progression (mean, SD, range)	11 (12, 9, 1–41)	
	Median interval to loss of ambulation (mean, SD, range)	22 (22, 11, 4–47)	
	Interval to loss of ambulation versus survival	$r^2 = 0.624$ ($p < 0.0005$)	
	K–M median	38 versus 20 ($p < 0.0005$)	
Specific symptoms	Median interval from loss of ambulation to death (mean, SD, range)	3 (7, 10, 0–42)	
	Had developed anarthria at census	57%	
	Median interval from onset to anarthria (mean, SD, range)	18 (21, 13, 0–77)	
	Interval to anarthria versus survival	$r^2 = 0.303$ ($p = 0.002$)	
	K–M median	56 versus 23 ($p = 0.011$)	
	Median interval from anarthria to death (mean, SD, range)	10 (12, 9, 2–43)	
	Reported emotionality	45%	
	Neurophysiology	80%	
Investigations	MRI	61%	
	CT only	8%	
	AChR antibodies	22%	
	Lumbar puncture	4%	
Interventions	Gastrostomy	78%	
	Median interval from onset to gastrostomy (mean, SD, range)	13 (15, 8, 1–42)	
	Median interval from diagnosis to gastrostomy (mean, SD, range)	3 (5, 6, 0–30)	
	Interval to gastrostomy versus onset to anarthria	$r^2 = 0.45$ ($p < 0.0005$)	
	Interval to gastrostomy versus onset to loss of ambulation	$r^2 = 0.44$ ($p < 0.0005$)	
	Median interval from gastrostomy to death (mean, SD, range)	10 (14, 11, 1–47)	
		(trend to longer survival in women, $p = 0.051$)	
	Riluzole use	43%	62%
	Riluzole use/not versus age of onset in years	64 versus 71, $p = 0.011$	

compare means (*T* test) of pertinent variables where they were common to both studies.

3. Results

A total of 49 patients with bulbar-onset were identified from 262 sporadic ALS cases recorded in the database. The results are summarised in Table 1.

3.1. Overall survival

Survival status at census was established for all subjects, with 82% deceased. Median survival from symptom onset for BO patients was 27 months (95% CI 20–34, range 6–84 months), and mean survival was 33 months (95% CI 27–39). For the previously published lower limb-onset dataset median survival was 41 months (95% CI 33–49, range 5–285 months), with a mean of 73 months (95% CI 53–92).

3.2. Demographics

The mean age of symptom onset for BO cases was 68 years (SD 11; range 43–90), significantly higher when compared to 60 years (SD 11; range 33–84) for the lower limb-onset dataset ($p < 0.0005$). Categorised age of onset was not related to survival by K–M analysis in BO patients.

Of the BO cases, 63% were female compared with 51% for the complete lower limb-onset dataset ($p < 0.0005$). Gender was unrelated to other variables or survival.

3.3. Diagnostic pathway

The mean time to diagnosis was 10 months (median 9, SD 6, range 2–34), compared to 22 months (median 15; SD 23, range 2–216) for the lower limb-onset dataset ($p < 0.0005$). This measure correlated only weakly with survival in linear regression ($r^2 = 0.144$, $p = 0.008$), but we observed a more significant effect on survival in K–M analysis categorised above and below the mean (median survival 33 versus 23 months, $p = 0.012$).

Prior referral to other specialist clinics was reported in 49% of BO patients, but this did not influence diagnostic latency or overall survival. Specialist clinics attended prior to neurology included otolaryngology in 54%, and stroke clinic in 42%, with one patient seen in both.

Neurologists referred 74% of cases to our tertiary clinic, stroke physicians 10%, GPs 14% and emergency medicine one case. This variable was not directly related to survival, but the mean diagnostic latency was longer when referral came via neurologists rather than others (11 versus 7 months, $p = 0.027$).

3.4. Progression beyond bulbar region

Symptomatic progression beyond the bulbar region at census was identified in 59% of cases, but was unrelated to other variables or survival. The median time from symptom onset to progression was 11 months (mean 12, SD 9, range 1–41), with equal initial progression to the upper and lower limbs. Neither initial site nor time interval to progression had any association with survival.

Loss of ambulation was recorded in all but two patients. The median time from symptom onset was 22 months (mean 22, SD 11, range 4–47). This time was significantly shorter for deceased patients (mean 20 versus 31 months, $p = 0.013$), strongly correlated with survival ($r^2 = 0.624$, $p < 0.0005$), and with a significant effect on survival in K–M analysis categorised above and below the mean (median survival 38 versus 20 months, $p < 0.0005$). The median survival from loss of ambulation for the deceased patients was 3 months (mean 7, SD 10, range 0–42).

3.5. Specific symptoms

At census 57% were recorded as having progressed to anarthria, with significantly more such patients having been referred to the stroke clinic (9/28 versus 1/21, $p = 0.03$). The median time from symptom onset to anarthria was 18 months (mean 21, SD 13, range 0–77). This was significantly shorter in the deceased patients (mean 13 versus 26 months, $p = 0.001$), with correlation to survival ($r^2 = 0.303$, $p = 0.002$), and a significant effect on survival in K–M analysis categorised above and below the mean (median survival 56 versus 23 months, $p = 0.011$). The median survival from anarthria to death was 10 months (mean 12, SD 9, range 2–43).

There was a very strong relationship between the time to development of anarthria and the time to loss of ambulation ($r^2 = 0.59$, $p < 0.0005$, Fig. 1). In contrast there was no significant correlation between the time to anarthria and time to progression of symptoms beyond the bulbar region ($r^2 = 0.11$, $p = 0.19$). Four cases (8%) held values below the mean for time to anarthria, and above the mean for time to loss of ambulation. The female:male ratio was 3:1.

Emotionality was reported in 45% of patients, though this was independent of the reporting of anarthria and other variables, and its occurrence had no influence on survival.

3.6. Investigations

MRI scan of the brain was performed in 61% of cases, with a further 8% undergoing solely CT brain. Significantly more patients reporting emotionality had undergone MRI scan of the brain during their diagnostic work-up (17/22 versus 13/27, $p = 0.045$). Neurophysiological studies were performed in 80% of patients, but there was no relationship to other variables. Acetylcholine receptor antibody testing was recorded in 22%. Lumbar puncture was recorded in two patients.

3.7. Interventions

Gastrostomy was performed in 78% of patients, but it was not linked to survival in K–M analysis. The median time from symptom

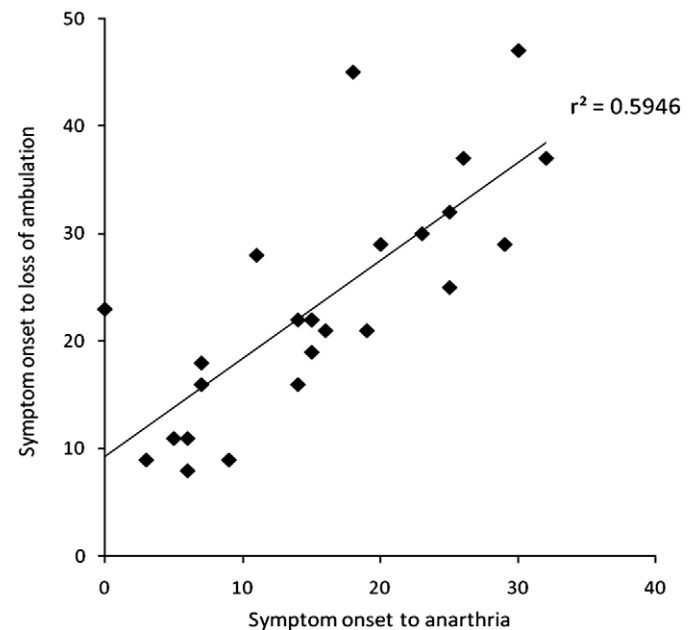


Fig. 1. Scatter-plot showing the strong correlation between the time from symptom onset to anarthria, and the time to loss of ambulation (in months). This has important implications for care planning when median survival after loss of ambulation in this series was only 3 months.

onset to gastrostomy was 13 months (mean 15, SD 8, range 1–42). The median time from diagnosis in our clinic to gastrostomy was 3 months (mean 5, SD 6, range 0–30; two patients were referred with gastrostomy *in situ*.) There was a correlation between time from symptom onset to gastrostomy, and both symptom onset to anarthria ($r^2 = 0.45$, $p < 0.0005$), and loss of ambulation ($r^2 = 0.44$, $p < 0.0005$). The median time from gastrostomy to death was 10 months (mean 14, SD 11, range 1–47), with a trend to longer survival in women (mean 16 versus 6 months, $p = 0.051$).

Riluzole use was recorded in 43% (62% in the lower limb-onset dataset), with a younger mean age at symptom onset in those BO patients who took the drug (64 versus 71 years, $p = 0.011$). There was no significant relationship to other variables including survival.

4. Discussion

This retrospective descriptive study of a group of bulbar-onset ALS patients confirms the long-recognised lower median survival overall, whilst specifically noting that nearly 10% of patients survived beyond 4 years from symptom onset (the upper limit of median survival in a review of prognostic factors in ALS [2]). We confirmed a higher proportion of females and higher mean age of symptom onset, and demonstrated that the development of anarthria is strongly predictive of the time to eventual loss of ambulation in this patient group, which may influence care planning. We noted a female bias (3:1) in the small sub-group of those with below-mean time (i.e. a relatively rapid course) to anarthria and above-mean time (i.e. a relatively delayed course) to loss of ambulation.

The progression of bulbar symptoms in those with limb-onset ALS has been identified as an independent prognostic factor [13]. Our study would support the notion that bulbar pathology *per se* is a major prognostic factor given that the time to progression (or not) beyond this territory did not appear to influence overall survival. The same study highlighted progression rate in the lower limbs as an independent prognostic factor in ALS [13]. We identified early loss of ambulation as a particularly adverse prognostic factor in BO patients (50% of patients dying within 3 months of this event), and to a lesser degree also the time interval to the development of anarthria. Both of these measures were also linked to earlier gastrostomy, and the continued uncertainty over which patients benefit most in terms of survival after PEG [2], suggests this warrants further study.

Unlike our findings in lower limb-onset ALS patients, the next symptomatic territory in BO ALS patients was split evenly between upper and lower limbs, rather than solely spreading to the upper limbs as one might expect in a simplistic contiguous model of distal spread of disease. Even acknowledging the concept of a complex summing 3D network of UMNs and LMNs [14], this contrast between 'proximal' (BO) and 'distal' (lower limb-onset) may hold important clues to focality and spread in ALS.

We observed the familiar concept of shortened diagnostic latency as a surrogate marker for more rapidly progressive cases (and so linked to reduced survival), and confirmed others' findings that this delay is shorter than in limb-onset patients [8,13]. We also demonstrated, perhaps surprisingly, that diagnostic latency was not independently compounded by inappropriate referral to other specialist clinics, to which nearly half the BO patients had been sent. It has been reported that 43% of BO patients had attended otolaryngology clinic and in nearly half of these a neuromuscular cause was missed by the specialist [15]. Our finding of a high frequency of referral to either stroke or otolaryngology clinics observed suggests that there may also be merit in targeted education of GPs about bulbar symptoms.

The high number of stroke clinic referrals is surprising given that the hallmark of vascular events is abrupt onset, rather than the invariably insidious nature of degenerative disorders. This may simply reflect the tendency for patients to attribute the onset of their

symptoms to a particular (often stressful) life event. Indeed, a single patient in our series reported complete anarthria as their first symptom to the GP and was referred to stroke clinic as a result (this female patient also went on to retain ambulation for nearly 2 years). More than one fifth of patients had serum tests for myasthenia gravis which may reflect the common reporting by ALS patients of variability in their symptoms during the early stages.

The apparent lengthening effect of diagnostic latency for patients referred by another neurologist may be artefactual. Such an effect, mediated according to whether a patient was referred to the neurologist by a GP or another specialist has been previously noted [8], and we were not able to ascertain this level of detail in our patients. Another reason may be that referral to a tertiary clinic by neurologists is driven by a desire to access multi-disciplinary care rather than for diagnostic purposes, and so may be initiated later in the disease course. Another source of delayed referral may be that non-specialist neurologists may place undue importance on EMG confirmation of widespread LMN involvement in BO cases, which may be absent in the early stages of the disease.

One third of patients did not have cerebral imaging recorded in our clinical notes, and we suspect under-ascertainment. Although rare case reports of mimics for bulbar-onset ALS exist [16], this mode of presentation rarely offers a tangible differential diagnosis to an experienced ALS neurologist, though a natural desire to exclude structural pathology persists (possibly driven in part by patient expectation).

Finally, we observed that a lower proportion of BO patients had taken riluzole (compared with lower limb-onset patients), and with a tendency for those doing so to be younger. Potential explanations might include a reluctance to prescribe or take the drug in those with swallowing difficulties, an avoidance of prescription in the older patient due to perceived co-morbidities or increased risk of side effects, or a greater motivation of younger patients to seek any disease prolonging treatment.

There are always marked limitations and need for caution in the interpretation of retrospective database analyses, given that they rely on uniformly rigorous note-keeping and access to such records, all of which can introduce unwanted bias, compounded by the relatively small size of our study group. Our observations that patients reporting emotionality were over-represented in those who had undergone MRI, and that those developing anarthria were over-represented in stroke clinics may be chance findings. We were not able to assess the timing of either symptom in relation to the actual investigation, and we are mindful of the inherent danger of multiple comparisons.

Many questions about BO ALS remain unanswered, but an important message for patients is the confirmation that heterogeneity of survival is still observed, in common with other presentations of ALS. We hypothesise that one aspect of the adverse prognosis, in addition to nutritional deficiency, may be earlier progression to brainstem respiratory centres due to their proximity to the pathologically affected areas associated with bulbar symptoms. Further prospective studies of ALS patients, focusing specifically on respiratory functional measures in conjunction with a more detailed study of the neuropathological correlates of respiratory failure are warranted.

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